Metabolic Modelling of *Aspergillus niger* and Grapevine Pathogenic Interactions

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Abstract. *Vitis vinifera* is threatened by phytopathogenic fungi such as *Aspergillus niger*, which infects berries from *véraison* through the post-harvest, compromising yield and food safety through mycotoxin production. The metabolic basis of this host–pathogen interaction remains largely unexplored. This project developed a reproducible *in silico* platform to characterise its metabolic interface. Two curated genome-scale metabolic models were integrated into a compartmentalised model linked by unidirectional transport reactions: iMS7199 for the host (green- and mature-berry variants) and iJB1325 for the pathogen. Infection was simulated by dynamic flux balance analysis using a phenomenological leakage model. In this model, host-to-pathogen nutrient flux scales with infection intensity and with host's cytoplasmic metabolic activity; infection intensity is a Michaelis–Menten function of fungal biomass. The mature berry is colonised earlier than the green berry ($f_{\text{infection}} = 0.95$ at 54 h vs. 70 h), with dominant leaked metabolites shifting from amino acids to organic acids and inorganic phosphate across *véraison*. In the mature-berry reconstruction the resveratrol-biosynthesis reaction carries no flux at the basal state and a quantitative growth–virulence trade-off was obtained for ochratoxin A and kotanin. Post-harvest infection is self-limiting through sucrose depletion. Single-gene deletion flagged 140 candidate fungal targets, but nomenclature incompatibility precluded host-orthologue screening, so their selectivity remains to be confirmed.

Keywords: FBA · dFBA · GSMMs · COBRApy · systems biology · plant–fungus interaction · mycotoxins · sustainable agriculture

1 Introduction

1.1 Background and Rationale

Viticulture is a cornerstone of Portugal's heritage [1]. Economically, the importance of *Vitis vinifera* (*V. vinifera*) cultivation extends beyond wine commercialisation, contributing to the national trade balance through exports and fostering regional development and growth in the tertiary sector through wine tourism [2].

Any factor threatening viticultural productivity or wine quality represents a challenge requiring mitigation strategies. Among the biotic threats are phytopathogenic fungi, which can reduce yields by causing vine diseases [3]. Although disease management has relied on chemical fungicides, concerns regarding their environmental impacts have intensified the search for sustainable alternatives [4]. Among these pathogens, *Aspergillus niger* (*A. niger*) is particularly relevant due to its detrimental effects on grape quality and its ability to produce mycotoxins that compromise food safety [5]. Despite this, the metabolic interaction between *A. niger* and *V. vinifera* remains poorly understood.

This project studies the metabolic interplay between these two species using Genome-Scale Metabolic Models (GSMMs). Constraint-Based Analysis (CBA) is employed to characterise the metabolic dependencies that sustain fungal infection and to identify essential reactions and genes that may represent targets for biocontrol strategies to a more resilient viticultural framework.

1.2 Objectives

The primary objective is to develop a computational framework for studying metabolic interactions between *V. vinifera* and *A. niger*. Consistent with its exploratory purpose, this study is organised around objectives rather than*a priori* hypotheses. To structure the approach, the following objectives were defined:

- **Curation and validation of GSMMs:** Perform targeted curation of the GSMMs of *V. vinifera* (iMS7199) [6] and *A. niger* (iJB1325) [7], checking stoichiometric consistency. Model reliability is assessed using MEMOTE [8].
- **Characterisation of interspecific nutrient cross-feeding:** Develop a simplified host–pathogen interaction model to identify the key metabolites exported by the host and used by *A. niger* for biomass production. Unidirectional transport constraints are introduced to quantify metabolic exchange.
- **Identification of fungal metabolic vulnerabilities:** Perform single-gene and -reaction deletion using COBRApy [9] to identify essential metabolic functions required for *A. niger* growth under host-simulated conditions, and screen the candidates against the host genome to assess their specificity.

The project was subsequently expanded to include: comparison between berry developmental stages, dynamic simulation, parametric robustness analysis, virulence characterisation and reconstruction of a complete infection trajectory.

2 State of the Art

Although *V. vinifera* was the first fruit species to have its genome sequenced, it retains structural complexity owing to pronounced heterozygosity, repetitive sequences and intraspecific variability [10,11]. *Véraison* is the most critical stage for modelling, as it marks a sharp metabolic transition — accumulation of soluble solids (sugars) and decrease in organic acids — that defines the nutritional landscape available to phytopathogens [12,13,14]. Berry metabolism is organised along two axes: primary metabolism (the source of carbohydrates, organic acids and amino acids, the main carbon and nitrogen substrates for the fungus) and secondary metabolism (bioactive defence compounds: phenolic compounds, stilbenes, and phytoalexins such as *trans*-resveratrol and ε-viniferin) [15,16,17].

A. niger is one of the most common and widely distributed species of its genus, recognised for its metabolic versatility and its role as an efficient "cellular factory" [18]. Its pathogenicity is associated with high-flux primary and secondary metabolic activity. Primary metabolism is characterised by elevated production of organic acids and the secretion of enzymes that facilitate the breakdown of host substrates. Secondary metabolism involves the biosynthesis of polyketides and alkaloids, including mycotoxins such as ochratoxin A (OTA), fumonisins and kotanin, which are associated with nephrotoxic and carcinogenic effects. These pathways constitute critical metabolic sinks that modulate energy homeostasis and contribute to virulence potential [19,20,21,22].

A. niger acts on *V. vinifera* mainly from ripening to harvest [23]. Infection is characterised by secretion of enzymes [24] and accumulation of organic acids, leading to tissue disintegration [25,26]. Although the grapevine activates immune responses (synthesis of stilbenes/phytoalexins) [27], the fungus often overcomes these barriers [28]. For constraint-based modelling, the critical features of this interaction are the infection-driven downregulation of glycolysis and the TCA cycle in the berry, compromising ATP production [29,30]; the increased sugar and reduced acidity with ripening, which favour fungal flux distribution [31]; and mycotoxin interference with host detoxification and redox balance [32,33].

CBA extends to the study of communities and interactions between species, but the literature is dominated by microbial systems and human pathogens [34]; plant–fungus interactions are poorly explored, and the *V. vinifera*–*A. niger* pair does not yet have an integrated GSMM. Compartmentalised integration, adopted here, merges the models and links them through transport reactions — transparent and easy to parameterise, but, with sequential Flux Balance Analysis (FBA), it does not arbitrate competition for resources simultaneously. For the temporal dimension, dynamic FBA (dFBA) [35] depends on kinetic parameters (Vmax, Km, uptake rates) that are often calibrated rather than experimentally measured, making robustness analysis a credibility requirement [36,37,38].

3 Materials and Methods

3.1 Study system and metabolic models

This study is entirely computational (*in silico*). *V. vinifera* and *A. niger* are represented by two public GSMMs. iJB1325 was obtained from the supplementary material of Brandl et al. (2018) [7]; iMS7199 models were downloaded from BioModels [39]. Two developmental variants of the host were used (green berry, pre-*véraison*; mature berry, post-*véraison*) to capture the metabolic reprogramming of ripening. Dimensions of the three models are summarised in Table 1.

3.2 Software and computational environment

The workflow was implemented in Python using COBRApy and Gurobi [40], with pandas [41] and NumPy [42] for data processing, Matplotlib [43] and

Seaborn [44] for visualisation, and MEMOTE for model quality assessment. Analyses were conducted in JupyterLab across nine sequential notebooks, using single-process optimisation due to model size.

Table 1. Dimensions of the three GSMMs used — number of reactions, metabolites and genes. iMS7199 is represented by two developmental-stage variants of the host (green, pre-véraison; mature, post-véraison); iJB1325 is the *A. niger* reconstruction.

Model	Organism	Reactions	Metabolites	Genes
iJB1325	*A. niger*	2320	1818	1325
iMS7199 (mature)	*V. vinifera*	4272	5143	7199
iMS7199 (green)	*V. vinifera*	4495	4399	6657

3.3 Model curation and validation

Each GSMM was audited and curated in a targeted manner, leaving growth unchanged. Gap-filling was deliberately not performed. Both reconstructions are published, growth-validated models, and the objective was to interrogate their existing metabolic structure, not to extend it. Curation was therefore restricted to targeted, traceable corrections that preserve the growth phenotype.

In iJB1325, ATP transport reaction `r2137` was constrained to unidirectional flux (lb $= 0$) to eliminate thermodynamically implausible back-flux. In iMS7199, reactions with IDs incompatible with the solver nomenclature were renamed, and compartments were remapped to semantic identifiers. Integrity was verified through round-trip validation (SBML export/re-import, $|\Delta\mu| < 10^{-6}$), and quality was assessed using MEMOTE. The full MEMOTE `test_consistency` suite proved computationally intractable for iMS7199 at this scale; per-reaction mass and charge balance was therefore checked directly with `check_mass_balance()`, returning a residual imbalance (`RXN0-5184__chlo`). The broader stoichiometric-consistency analysis could not be run at this scale and remains a limitation.

3.4 Construction of the compartmentalised host–pathogen model

Host and pathogen were prefixed (`vv_` / `an_`) and merged into a single model, linked by an infection compartment (`inf`) with unidirectional nutrient flux:

$$\text{Xylem/Phloem} \rightarrow \text{[berry cytosol]} \xrightarrow{T1} \text{[inf]} \xrightarrow{T2} \text{[fungal extracellular]}$$

Grape tissue receives nutrients only from xylem/phloem, fungus imports exclusively via T2 (direct `an_BOUNDARY_*` uptakes blocked), T1 and T2 are strictly unidirectional (lb $= 0$), chloroplast metabolites do not feed T1 and charged forms and ammonium were excluded to avoid LP instability. 19 interface metabolites were defined (listed in Table 2), with IDs verified in both models. Merging the two reconstructions and adding the 19 T1 and 19 T2 transports yields 6630 reactions. The original reconstructions already contained 994 orphan metabolites (species not consumed or produced by any reaction); these were removed during merging and, adding the 19 interface metabolites, the mature community model contains 5986 metabolites.

3.5 Dynamic infection simulation (dFBA) and leakage model

Temporal progression was simulated by dFBA: although host and pathogen share a single merged model, at each time step their objectives are optimised *separately* (sequential FBA, not a single joint LP) — the host supplying, through the leakage bounds, the nutrients available to the pathogen — and the state variables are then updated [37,38]. Coupling was represented by a phenomenological leakage model that does not impose external concentrations. Infection intensity follows Michaelis–Menten (MM) kinetics of the fungal biomass:

$$f_{\text{infection}} = f_{\max} \frac{A_{\text{niger}}}{A_{\text{niger}} + \text{half}_{\text{sat}}}, \qquad f_{\max} = 1, \tag{1}$$

where $f_{\max} = 1$ is the maximal (fully saturated) infection intensity, A_{niger} and half_{sat} are in gDW/L (so $f_{\text{infection}}$ is dimensionless), and half_{sat} is the fungal biomass at which $f_{\text{infection}} = f_{\max}/2 = 0.5$. The upper bound of each T1 transport reaction is proportional to the infection intensity and to the metabolic activity of the berry:

$$\text{T1}_{\text{ub}}[\text{met}] = f_{\text{infection}} \cdot \text{proxy}_{\text{met}}, \qquad \text{proxy}_{\text{met}} = \sum_r |v_r|, \tag{2}$$

where the sum runs over the cytosolic reactions r that consume the metabolite, so the proxy reflects the host's internal turnover (availability) of each metabolite, not its uptake from xylem/phloem. Two quantities must be distinguished: the *leakage* is the realised host-to-pathogen nutrient flux carried by a T1 transport at a given time, whereas the *leakage proxy* ($\text{proxy}_{\text{met}}$) is a fixed per-metabolite *capacity* — the host's internal cytosolic turnover of that metabolite — obtained once by parsimonious FBA (pFBA) [45,46] of the berry at maximum capacity and held constant. The two are linked by $T1_{\text{ub}}[\text{met}] = f_{\text{infection}} \cdot \text{proxy}_{\text{met}}$: the proxy is the ceiling on how much of a metabolite can leak, and the realised leakage is that ceiling scaled by the infection intensity. The *total cytoplasmic leakage proxy* quoted in the Results is the sum of this proxy over all interface metabolites, i.e. the berry's total leakage *capacity*. The host's growth capacity (the upper bound on its biomass-production rate) is constrained by

$$f_{\text{host}} = 1 - f_{\text{infection}}, \qquad \mu_{\text{host}}^{\text{ub}} = f_{\text{host}} \cdot \mu_{\max}. \tag{3}$$

f_{host} is a phenomenological choice, not a mechanistically derived relationship: it is the simplest decreasing function mapping $f_{\text{infection}} \in [0, 1]$ onto a host-capacity fraction without additional free parameters, and is not claimed to capture the true kinetics of host suppression. The fungal biomass is updated with a first-order death rate and a ceiling at the carrying capacity:

$$A_{\text{niger}}(t + \Delta t) = \min(A_{\text{niger}} + (\mu_{\text{An}} - d)\, A_{\text{niger}}\, \Delta t,\ K_{\text{fungus}})\,. \tag{4}$$

While $A_{\text{niger}} \ll K_{\text{fungus}}$ this hard cap coincides with a logistic term, differing only near the plateau; the logistic form is retained for the post-harvest phase, where the smooth deceleration drives the collapse. A carrying capacity is imposed because berry substrate and space are finite; in the post-harvest phase it contracts with the depleting sucrose reserve, driving the collapse.

Baseline parameters: $\Delta t = 0.5$–1.0 h, $K_{\text{fungus}} = 50\,\text{gDW/L}$, $\text{half}_{\text{sat}} = 0.01\,\text{gDW/L}$. K_{fungus} is a generous ceiling consistent with reported *A. niger* biomass densities [47], and half_{sat} is set at the scale of a typical initial inoculum, so that the infection intensity rises steeply only once the fungus has established; because both K_{fungus} and half_{sat} are phenomenological, their influence on the dynamics is characterised explicitly in the robustness analysis. In the full trajectory (green $\rightarrow$ mature $\rightarrow$ post-harvest, 432 h), stage-specific leakage factors (20% $\rightarrow$ 100% $\rightarrow$ 100%) and phase-specific first-order death rates (0.003, 0.002 and $0.050\,\text{h}^{-1}$ for the green, mature and post-harvest phases) were applied; the death rate is low while the substrate remains favourable (green and mature) and an order of magnitude higher post-harvest, when sucrose exhaustion renders the tissue hostile and drives the collapse. Glucose/fructose T1 flux was scaled by the dimensionless MM saturation factor of the sucrose reserve, $\text{mm}_{\text{suc}} = \frac{c_{\text{suc}}}{c_{\text{suc}}+K_M} \in [0,1]$ (with $K_M = 10\,\text{mmol}\,\text{L}^{-1}$), which modulates the maximum given by the unscaled flux it multiplies [48,49,50,51]. In the post-harvest phase the carrying capacity itself contracts with the sucrose reserve, $K_{\text{eff}} = \max(0.01,\, K_{\text{fungus}} \cdot \text{mm}_{\text{suc}})$, and the biomass follows a logistic update, which drives the collapse as the reserve is exhausted:

$$A_{\text{niger}}(t+\Delta t) = A_{\text{niger}} + \left[\mu_{\text{An}}\,(1 - A_{\text{niger}}/K_{\text{eff}}) - d\right] A_{\text{niger}}\,\Delta t. \tag{5}$$

3.6 Essentiality, virulence and statistical analyses

Each interface transport was individually inactivated (knockout) to identify essential metabolites; minimal medium of the isolated fungus was determined with `minimal_medium()` and gene essentiality was assessed by single-gene deletion. Mycotoxin capacity was obtained by switching the objective function to toxin secretion; the growth–virulence trade-off was quantified by forcing increasing secretion rates and measuring the growth penalty. The dose–response relationships (glucose/sucrose/carbon–biomass) were characterised by linear regression.

The robustness of dFBA was assessed by a 2D parameter scan of $\text{half}_{\text{sat}} \times K_{\text{fungus}}$ — two parameters without an experimental counterpart, governing infection timing and biomass magnitude respectively — complemented by one-at-a-time profiles. Marginal sensitivity was quantified by elasticity — the percentage change in an output relative to the percentage change in the perturbed parameter — and the spread of the results by the coefficient of variation. The primary robustness metric was t_{f95} (time to $f_{\text{infection}} = 0.95$, equivalent to $A_{\text{niger}} = 19 \times \text{half}_{\text{sat}}$); since $19 \times \text{half}_{\text{sat}} \ll K_{\text{fungus}}$, it is reached during exponential growth and is therefore independent of K, isolating the establishment dynamics (set by half_{sat}) from the biomass ceiling (set by K). LP shadow prices and reduced costs were used for local sensitivity analysis [52,53,54]. Interface fluxes were visualised using a pathway map generated programmatically (available in this project repository). As a complementary, fully coupled check, a tipping-point analysis was performed. With T1/T2 left open, the host biomass was forced to a series of fixed growth rates and fungal growth was maximised within a single joint optimisation, identifying the host activity above which the fungus can no longer be sustained.

4 Results

4.1 Model curation and characterisation

With the models' exchange reactions left at their default uptake bounds, iJB1325 (*A. niger*) reaches a glucose-limited $\mu \approx 0.94\,\mathrm{h}^{-1}$, while the two host variants attain $\mu = 0.1541\,\mathrm{h}^{-1}$ (mature) and $\mu = 0.1446\,\mathrm{h}^{-1}$ (green). Curation preserved each model's growth phenotype (the predicted growth rate was identical before and after curation); a single residual mass-balance error persists in the mature reconstruction (`RXN0-5184__chlo`). MEMOTE returns an aggregate quality score (0–100%) combining stoichiometric consistency with annotation coverage; it assigned total scores of 29.1% (iJB1325) and 4.0% (both iMS7199 variants). The low host scores stem almost entirely from absent metabolite annotation — reaction-level annotation (60%) is comparable to iJB1325's (53.4%) — and are therefore an annotation artefact rather than a reflection of stoichiometric quality.

Comparing gene essentiality between the two developmental stages (green vs. mature berry), of the 74 genes essential in the mature berry 73 are also essential in the green, and the single gene that gains essentiality upon ripening is `Vitvi02g00435_t001` (anthocyanidin synthase, ANS/LDOX), which catalyses RXN-7785 (its knockout reduces μ to zero in the mature berry). Full per-model characterisation is provided in the repository.

4.2 Mature host-fungus dFBA

Table 2. Cytoplasmic leakage proxy (T1 flux, mmol/gDW/h) of the interface metabolites in the green and mature berry, from a representative pFBA solution. The three remaining interface species (O_2, CO_2, water) are gases/solvent and are excluded from the leakage proxy.

Metabolite	Green	Mature	Metabolite	Green	Mature
Citrate	1.20	19.37	Leucine	0.10	0.15
Phosphate	2.03	13.34	Fumarate	0.05	0.08
Glycine	0.56	2.59	Proline	7.98	0.07
Alanine	0.09	1.95	Succinate	0.04	0.05
Glutamate	18.15	1.54	Sulfate	0.01	0.02
Nitrate	0.79	1.46	Adenine	0.02	0.01
Serine	0.06	1.28	Glucose	0.00	0.00
Malate	1.63	0.19	Fructose	0.00	0.00

In the coupled host–pathogen model, the berry maintains $\mu = 0.1541\,\mathrm{h}^{-1}$; the fungus cannot grow with T2 blocked ($\mu = 0$) but reaches $\mu = 0.1809\,\mathrm{h}^{-1}$ with T1/T2 open, confirming its dependence on host leakage. The cytoplasmic proxy is dominated by citrate and phosphate (Table 2).

Sweeping the infection intensity gives a continuous (non-switching) fungal response, saturating at $f_{\mathrm{infection}} \approx 0.55$ ($\mu_{\mathrm{An}} \approx 0.175$); because the leakage uses the static proxy, the accompanying host decline is imposed by the f_{host} cap rather than emergent competition. Tipping-point analysis confirms genuine competition for the host cytosolic pool only when the host is driven to essentially its own maximum ($\mu_{\mathrm{host}} \approx \mu_{\mathrm{max}}$). In the dFBA, *A. niger* saturates at the imposed $K = 50$ gDW/L by $t \approx 85$ h ($f_{\mathrm{infection}} = 0.9998$ at 120 h).

Of the 19 interface metabolites, single-transport knockout flags only O_2 and sulfate as essential: glucose and nitrate are individually dispensable, since the open interface supplies alternative carbon and nitrogen sources. Accordingly, the `minimal_medium()` for the isolated fungus is four-component (glucose, nitrate, O_2, sulfate). Single-gene deletion flagged 140 essential *A. niger* genes; selectivity could not be assessed.

4.3 Infection: green vs. mature berry

Colonisation — defined here as the infection intensity reaching $f_{\mathrm{infection}} = 0.95$ (time t_{f95}) — occurs 16 h earlier in the mature berry ($t_{f95} = 54$ h vs. 70 h; Fig. 1), and the fungal biomass likewise peaks earlier (time to peak reduced by 15.5% relative to the green berry). This advance tracks the mature berry's 28.7% higher total cytoplasmic leakage proxy, i.e. its greater total leakage capacity (Table 2). The dominant leaked metabolites shift from glutamate and proline (green) to citrate and phosphate (mature berry), with glucose absent from both. At saturation the two developmental stages (green and mature) reach a near-identical fungal growth rate, and the carrying capacity is identical by construction.

4.4 Parametric robustness (dFBA)

The robustness scan tested 49 parameter combinations (7 values of $\mathrm{half}_{\mathrm{sat}} \times 7$ of K_{fungus}) and split into two regimes. In the biologically plausible regime — the 28 of 49 combinations with $\mathrm{half}_{\mathrm{sat}} \leq 0.025$ gDW/L— infection always ran to completion, the final infection intensity being effectively invariant ($f_{\mathrm{infection}}$ between 0.995 and 1.000) owing to the self-reinforcing leakage–biomass coupling. In the remaining 21 combinations ($\mathrm{half}_{\mathrm{sat}} \geq 0.050$ gDW/L) the initial infection intensity was too low for the fungus to establish, so it did not grow; this "circular-dependency" regime is a structural artefact of the MM formulation and was excluded. Within the plausible regime the time to colonisation t_{f95} varied with $\mathrm{half}_{\mathrm{sat}}$ (18–102 h) but was independent of K_{fungus}, whereas the fungal biomass peak scaled linearly with K_{fungus} (a 1% change in K giving a 1% change in the peak, i.e. unit elasticity) and was insensitive to $\mathrm{half}_{\mathrm{sat}}$.

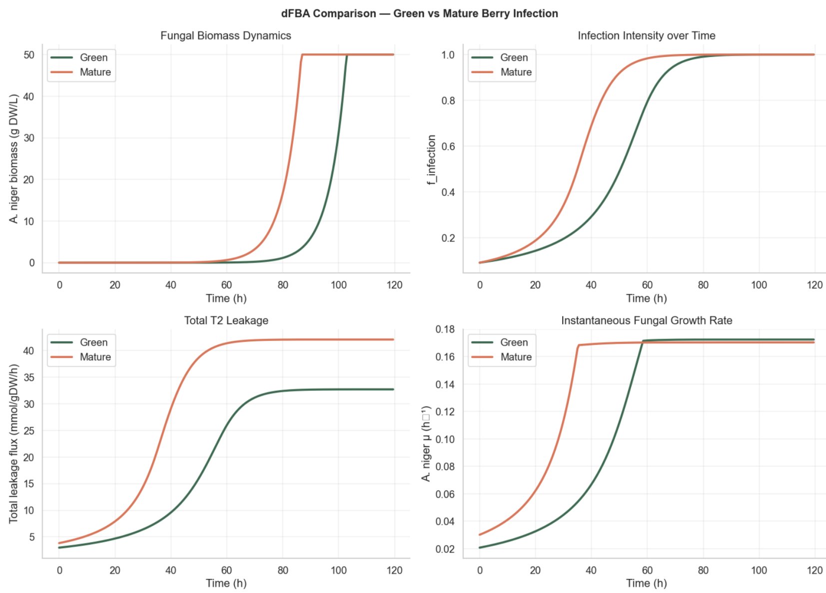

Fig. 1. Stage-resolved infection dynamics, green vs. mature berry. Time courses obtained by dFBA under identical kinetic parameters, so that the only difference between the two stages is each berry's cytoplasmic leakage proxy. Each panel plots, against simulated time: fungal biomass; infection intensity $f_{\mathrm{infection}}$; total host-to-pathogen leakage (summed T2 flux); and instantaneous fungal growth rate, i.e. the specific growth rate μ_{An} returned by the FBA solved at each time step. The mature berry reaches $f_{\mathrm{infection}} = 0.95$ at 54 h (vs. 70 h for the green) and its biomass peaks 16 h earlier, driven by its 28.7% higher total leakage capacity (cytoplasmic leakage).

4.5 Virulence: mycotoxins and host defence

iJB1325 contains complete pathways for OTA and kotanin (fumonisins are orphan entities). No mycotoxin is produced at maximum growth; when secretion is forced, growth declines linearly, with a steeper penalty per unit for kotanin than OTA (47.4% vs. 36.5%). Under the infection medium, OTA capacity falls from 2.7273 to 0.6818 mmol/gDW/h (25% of the synthetic value).

Incremental nutrient analysis identifies phosphate as the dominant growth-limiting nutrient: its addition alone matches all other nutrients combined, and the complete mixture supports 19% higher growth than the synthetic medium. Of the 156 host defence reactions, 32 carry flux at baseline and 31 decrease under infection; resveratrol biosynthesis (RXN-87) carries no flux at the basal state.

4.6 Complete infection time course

A single continuous 432 h dFBA simulation was run across the three developmental phases (green → mature → post-harvest). At the *véraison* transition (48 h) the T1 leakage increases about five-fold, triggering exponential fungal growth that reaches the carrying capacity K within roughly 45 h; in the post-harvest phase the sucrose reserve is depleted and the fungal biomass collapses to near zero by day 18. This simulation couples all five state variables of the model — fungal biomass, the sucrose-driven carrying capacity, infection intensity, T1 leakage and mortality — in one run. The complete time course is shown in Fig. 2.

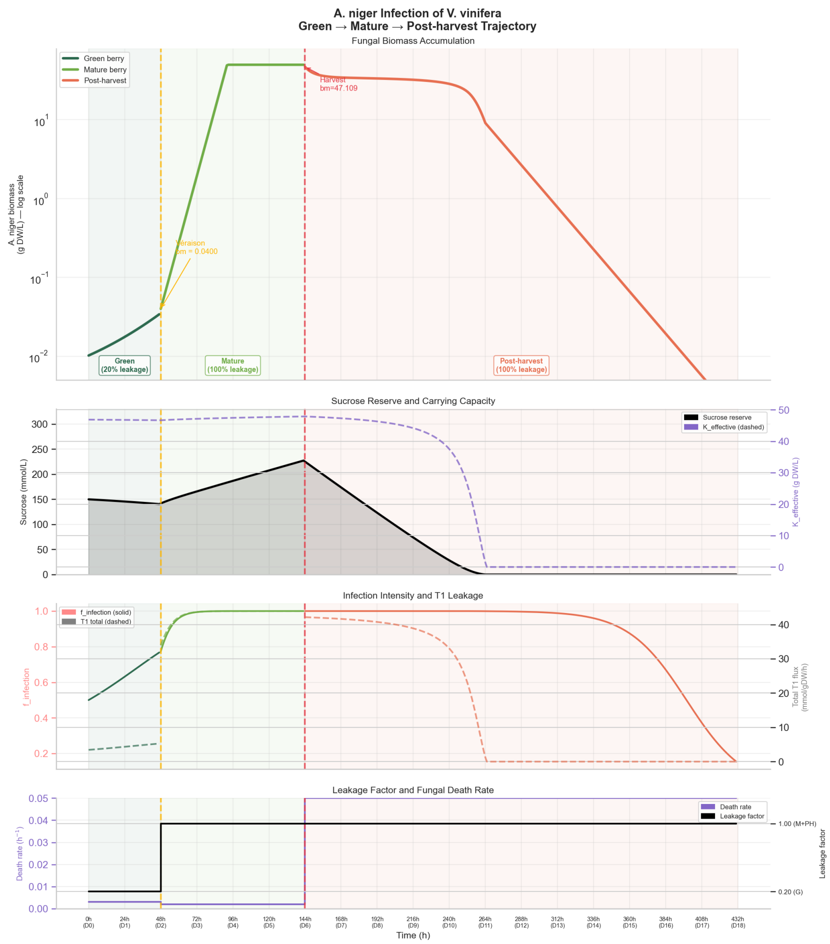

Fig. 2. Complete time course of the infection model. *A. niger* infection along green → mature → post-harvest (18 days): fungal biomass; sucrose reserve and carrying capacity; infection intensity and T1 leakage; fungal leakage factor and mortality rate.

5 Discussion

5.1 *Véraison* as the susceptibility turning point

The central result is that, in the stage-matched comparison (Fig. 1), the mature berry is colonised earlier and faster than the green. Because both stages were simulated with identical kinetic parameters, the only stage-specific input is each berry's cytoplasmic leakage proxy — a property of its own metabolic network — so this advance is attributable to the berries' metabolic state rather than to any imposed stage-specific parameter. This agrees with established knowledge — black *A. niger* rot predominantly affects mature and overripe berries, rarely damaging green, pre-*véraison* tissue [23,55,56]. The complete infection time course adds a complementary, empirically motivated layer: a leakage factor ramping from 20% (green) to 100% (mature) at *véraison*, amplifying the T1 flux ∼5-fold. *Véraison* thus emerges as the intervention window of greatest potential impact for biocontrol.

5.2 Metabolic signature of the interface and the role of phosphate

The shift in the dominant leakage metabolite — glutamate to citrate — mirrors the transition from the amino-acid catabolism of dividing tissue [57,58,59] to the TCA-centric, organic-acid metabolism of the ripening berry [60,61,62]: the metabolic character of infection differs by stage even when the carrying capacity is unchanged. Counter-intuitively, glucose is absent from the interface (proxy = 0); in this model the carbon driving fungal growth is citrate and amino acids, not free cytoplasmic glucose [63], because iMS7199 routes carbon through sucrose and anomeric forms rather than the free glucose pool.

Incremental analysis identifies phosphate as the dominant growth-limiting nutrient [64,65]: under infection *A. niger* is not carbon- but phosphorus-limited — a candidate bottleneck warranting experimental validation.

5.3 Host defence at the basal state and the virulence trade-off

That the anthocyanidin synthase gene (ANS/LDOX [66]) gains essentiality only in the mature berry shows the model recapitulating a hallmark molecular event of *véraison*. Its downstream consequence — RXN-87 (resveratrol biosynthesis) carrying zero flux and 31 of 32 active defence reactions decreasing under host suppression — reflects the basal metabolic state encoded in iMS7199, not a prediction that resveratrol cannot be induced *in planta* (the model omits infection-induced transcriptional reprogramming, and stilbene synthases are strongly induced upon *Aspergillus* challenge [67,68]). The model therefore localises a potential vulnerability in the stilbene branch at the basal state without establishing that defence is compromised during infection.

As for virulence, no mycotoxin is produced at maximum growth, consistent with its induction by nitrogen or carbon limitation [69,70]. The growth–virulence trade-off has a structural basis — kotanin requires two coupled polyketide synthases, OTA a single PKS–amino-acid assembly [71,72] — and the fall in OTA capacity to 25% under infection suggests the carbon-rich infection environment is limited in specific toxin precursors.

5.4 Model robustness and validity

The $\mathrm{half_{sat}} \times K_{\mathrm{fungus}}$ scan probes the model's dynamic behaviour, not its structure: within the plausible regime t_{f95} depends on $\mathrm{half_{sat}}$ (whose calibration would need fungal-biomass time series) while the peak scales linearly with K. The headline qualitative features are therefore structural rather than stress-tested — the invariance of the final intensity ($f_{\text{infection}} \to 1$), the non-switching gradient and the citrate/phosphate signature all follow from the self-reinforcing leakage–biomass coupling and the static proxy. The proxy is not strictly unique (pFBA alternate optima give, e.g., phosphate 13.27–13.34 mmol/gDW/h) and would benefit from deterministic tie-breaking or an FVA-based definition; the circular-dependency regime ($\text{half}_{\text{sat}} \geq 0.050$) is likewise a methodological artefact of MM leakage formulations [73,74].

6 Conclusion

This project built a reproducible *in silico* platform — two curated GSMMs integrated into a compartmentalised model and simulated by FBA and dFBA with a phenomenological leakage model — to dissect the *V. vinifera*–*A. niger* metabolic interface. In a stage-matched comparison the mature berry was colonised earlier than the green with no stage-specific virulence parameter; because this difference is driven by the mature berry's higher cytoplasmic leakage proxy (a model input), it should be read as consistency with the known *véraison* susceptibility window rather than independent validation, and it identifies *véraison* as the most promising intervention window. The complete infection time course adds an empirically motivated leakage factor that intensifies at *véraison*.

The dominant leaked metabolites differ between developmental stages — amino acids in the green berry, organic acids and phosphate in the mature — while the infection endpoint (saturation level and carrying capacity) is unchanged; this points to phosphate, not carbon, as the candidate growth-limiting bottleneck for the fungus. The essentiality and virulence analyses yield candidate intervention points, including a quantifiable growth–virulence trade-off, but their translational value is conditional: the resveratrol branch carries no flux at the basal state, and the 140 essential fungal genes cannot yet be claimed as selective targets, as host-orthologue screening was precluded by nomenclature incompatibility; the third objective was thus only partially met.

These results are bounded by the platform's methodological assumptions: host and fungus are optimised sequentially rather than in joint competition, and the leakage relies on a static pFBA proxy with phenomenological kinetic parameters (leakage factors, death rates, phase durations) and a constant, well-mixed host biomass, so the host trajectory is qualitative and spatial and regulatory dynamics — including infection-induced transcriptional reprogramming — are not captured. Beyond these, NCBI–BioCyc nomenclature incompatibility precluded host-orthologue screening and the fumonisin pathway is absent from iJB1325. Accordingly, the priority steps are to harmonise the gene nomenclature to validate the specificity of the 140 essential targets, to calibrate half_{sat} and the death rates against fungal-biomass time series, and to experimentally validate the hubs flagged by the shadow-price and essentiality analyses — the route to turning the candidate targets into validated ones and consolidating the platform as a theoretical basis and metabolic map for sustainable biocontrol.

Code and data availability

All code (nine notebooks), curated SBML models (iJB1325 and the green/mature iMS7199 variants), coupled host–pathogen model, 19 interface metabolites with their identifiers in both models and pinned environment (`requirements.txt`) are available at `https://github.com/PG58816/aspergillus-vinifera-gem`. The repository maps notebooks to the figures and results reported here.

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